AWS S3 EVENT-DRIVEN AUTOMATION USING LAMBDA

Project Overview

This project demonstrates how to implement event-driven automation using Amazon Simple Storage Service (Amazon S3) and AWS Lambda. It covers creating an S3 bucket, developing a Lambda function, configuring an S3 event notification to trigger the function when a file is uploaded, and verifying the automatic execution through Amazon CloudWatch logs.

Technology Used

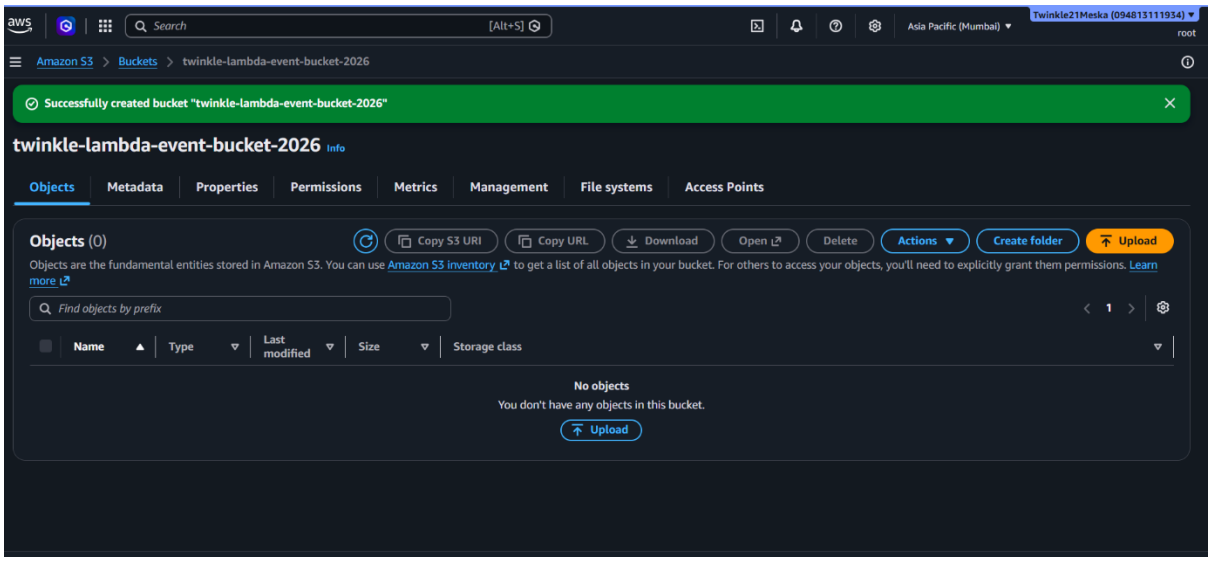
- Amazon Web Services (AWS)
- Amazon S3
- AWS Lambda
- Python
- Amazon CloudWatch
- S3 Event Notifications and IAM Permissions

AWS Region: Asia Pacific (Mumbai) — ap-south-1

Prepared by: Twinkle Meska

1. Amazon S3 Bucket Creation

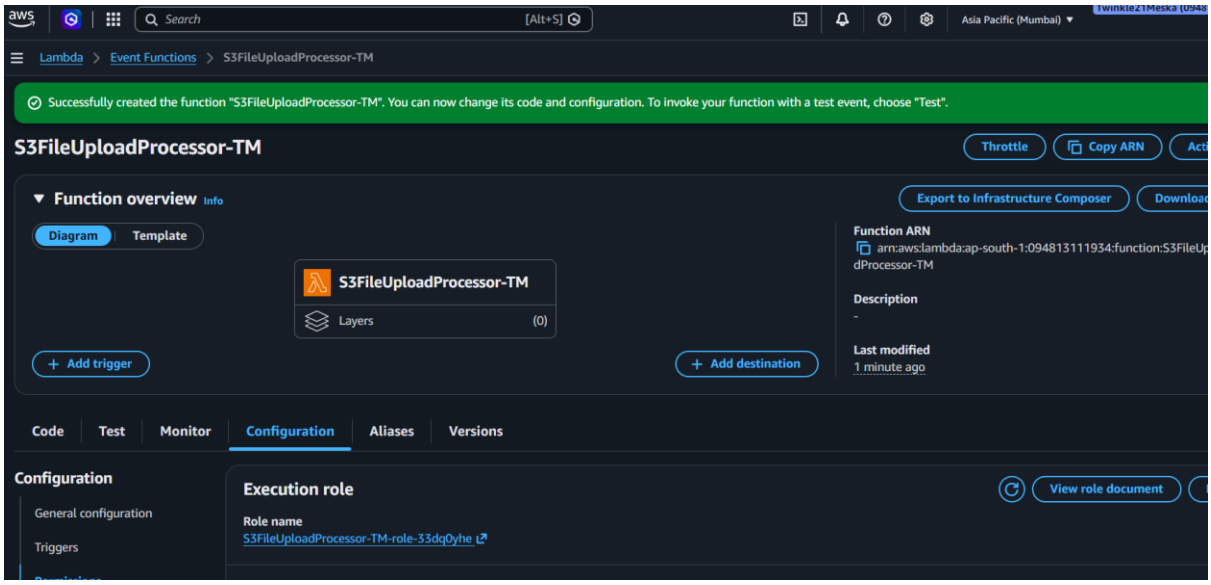
The first step in implementing event-driven automation is to create an Amazon S3 bucket. An S3 bucket is a cloud storage container used to store objects such as CSV files, text files, and other data. In this project, a dedicated bucket named twinkle-lambda-event-bucket-2026 is created to store input files. When a new file is uploaded to this bucket, an S3 event notification will be configured to trigger an AWS Lambda function automatically.



2. AWS Lambda Function Creation

AWS Lambda is a serverless computing service that runs code in response to events without requiring the user to manage servers. In this project, a Lambda function named S3FileUploadProcessor is created using Python. The function will be configured to respond to file-upload events from the Amazon S3 bucket and record details of the uploaded file in Amazon CloudWatch logs.

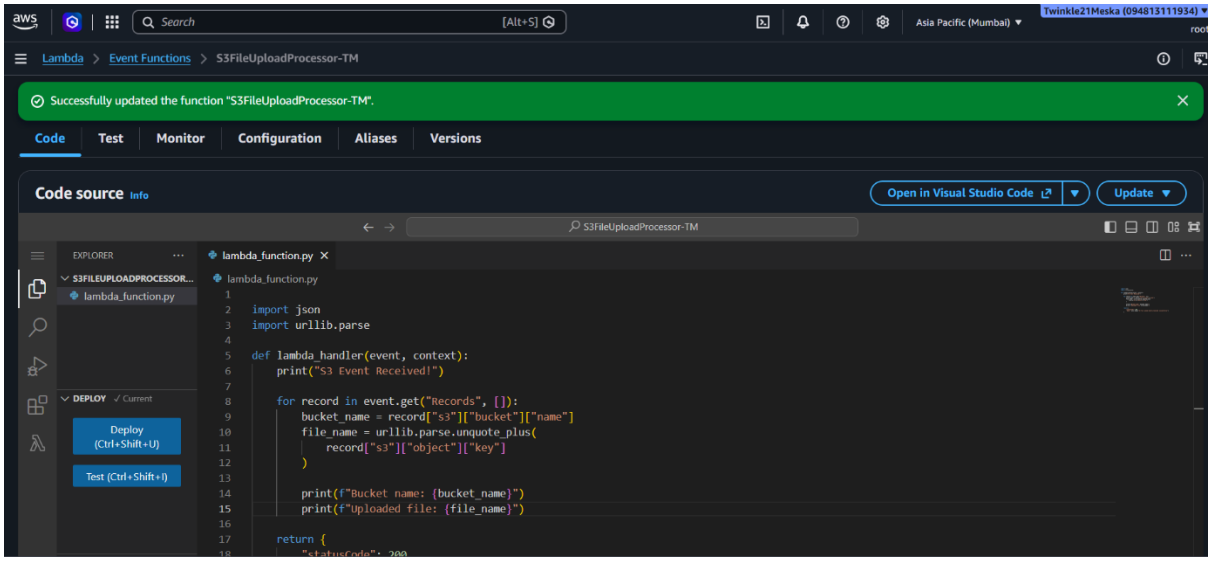
Creation of the AWS Lambda function for processing S3 file-upload events.



3. Lambda Function Code Implementation

Python code was implemented in the AWS Lambda function to process Amazon S3 event notifications. When an S3 object is uploaded, the function extracts the bucket name and uploaded file name from the event record. The details are printed to the function logs, which can be viewed in Amazon CloudWatch.

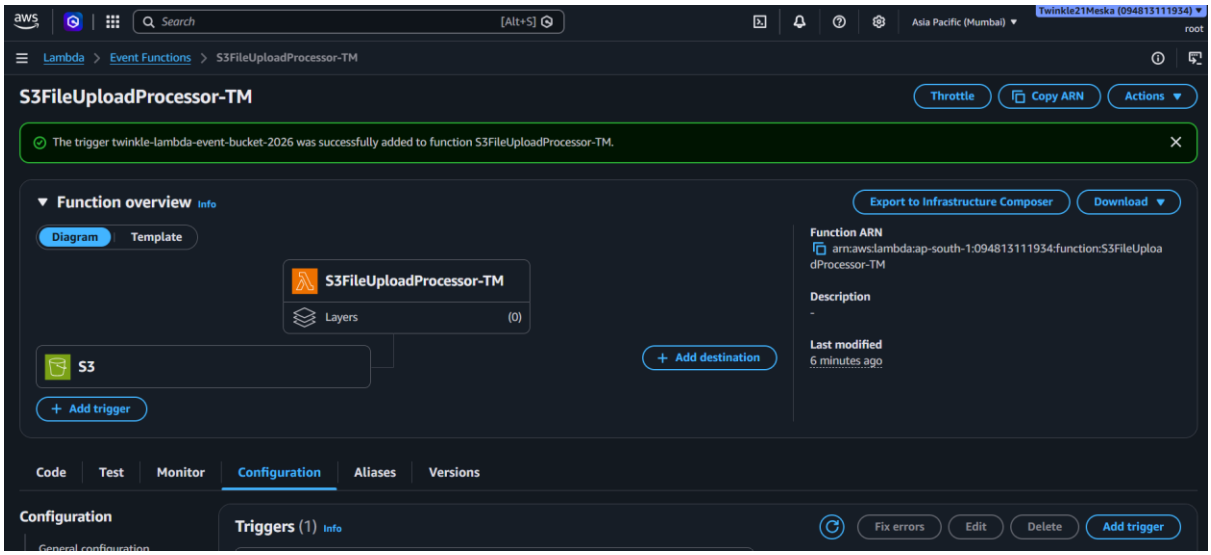
Python code implementation for processing S3 file-upload events.



4. Amazon S3 Trigger Configuration

An Amazon S3 trigger was configured for the AWS Lambda function to enable event-driven automation. The trigger listens for object creation events in the bucket twinkle-lambda-event-bucket-2026. When a new file is uploaded, Amazon S3 invokes the S3FileUploadProcessor-TM Lambda function, allowing it to extract and log the bucket name and uploaded file name.

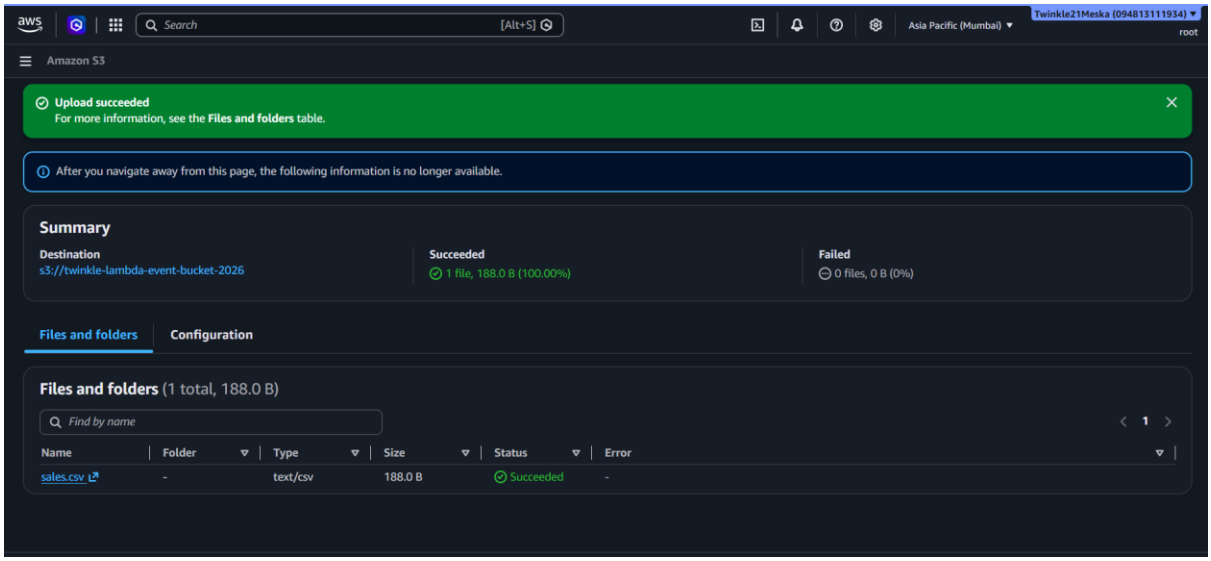
Configuring Amazon S3 to trigger AWS Lambda on object creation events.



5. Sample CSV File Upload to Amazon S3

A sample CSV file named sales.csv was created with employee details and uploaded to the Amazon S3 bucket twinkle-lambda-event-bucket-2026. This file serves as the input object for testing event-driven automation. When the object is created in the bucket, the configured S3 event notification should invoke the AWS Lambda function automatically.

Uploading a sample CSV file to Amazon S3 for event-driven automation testing.



6. Verification of Event-Driven Automation

The event-driven workflow was tested by uploading sales.csv to the configured Amazon S3 bucket. The Lambda function logs were checked in Amazon CloudWatch to verify whether the S3 object creation event invoked the function and whether the bucket name and uploaded file name were recorded successfully.

Verification of S3-triggered Lambda execution using Amazon CloudWatch logs.

